

1 Osnove teorije

1.1 BCS teorija pri končni temperaturi

BCS model, kjer je $\xi_{\mathbf{k},\sigma} = \epsilon_{\mathbf{k},\sigma} - E_F$ in $k := (\mathbf{k}, \uparrow)$, $-k := (-\mathbf{k}, \downarrow)$:

$$H = \sum_k \xi_k (c_k^\dagger c_k + c_{-k}^\dagger c_{-k}) - V \sum_{k,k'} c_k^\dagger c_{-k}^\dagger c_{-k'} c_{k'} \quad (1)$$

Naredimo približek povprečnega polja, definiramo $b_k = c_k c_{-k}$ in $b_k = \langle b_k \rangle + \delta b_k = F_k + \delta b_k$, sledi (do $\mathcal{O}(\delta b_k^2)$)

$$b_k^\dagger b_{k'} \simeq F_k^* F_{k'} + F_k^* \delta b_{k'} + F_{k'} \delta b_k^\dagger = F_k^* b_{k'} + F_{k'} b_k^\dagger - F_k^* F_{k'} = \langle b_k^\dagger \rangle b_{k'} + b_k^\dagger \langle b_{k'} \rangle - \langle b_k^\dagger \rangle \langle b_{k'} \rangle \quad (2)$$

Iz česar sledi:

$$H = \sum_k (c_k^\dagger \quad c_{-k}) \begin{pmatrix} \xi_k & -\Delta_k \\ -\Delta_k^* & -\xi_k \end{pmatrix} \begin{pmatrix} c_k \\ c_{-k}^\dagger \end{pmatrix} + \frac{|\Delta|^2}{V} + \sum_k \xi_k \quad (3)$$

Ob definiciji $\Delta_k = \sum_{k'} V(k - k') \langle c_{k'} c_{-k'} \rangle$, vzamemo še lokalno interakcijo, zato je $V(q) \simeq V$ in $\Delta_k \simeq \Delta$. Naredimo Bogoljubov rotacijo, privzamemo $u_k, v_k \in \mathbb{R}$ in simetrijo okoli E_F : $u_k = u_{-k}$ ter $v_k = v_{-k}$. Iz CCR sledi $u_k^2 + v_k^2 = 1$. Kvazidelci so definirani z rotacijo

$$\begin{pmatrix} c_k \\ c_{-k}^\dagger \end{pmatrix} = \begin{pmatrix} u_k & v_k \\ -v_k & u_k \end{pmatrix} \begin{pmatrix} \alpha_k \\ \alpha_{-k}^\dagger \end{pmatrix} \quad (4)$$

nov Hamiltonjan pa postane

$$H = \sum_k (\alpha_k^\dagger \quad \alpha_{-k}) \begin{pmatrix} \xi_k u_k^2 + u_k v_k (\Delta_k + \Delta_k^*) - \xi_k v_k^2 & -2\xi_k u_k v_k + \Delta_k u_k^2 - \Delta_k^* v_k^2 \\ -2\xi_k u_k v_k - \Delta_k v_k^2 + \Delta_k^* u_k^2 & \xi_k v_k^2 - \xi_k u_k^2 + u_k v_k (\Delta_k + \Delta_k^*) \end{pmatrix} \begin{pmatrix} \alpha_k \\ \alpha_{-k}^\dagger \end{pmatrix}, \quad (5)$$

kjer zahtevamo, da so izvendiagonalni členi enaki 0. Iz CCR zahteve lahko vzamemo parameterizacijo (tu sem vzel kompleksno nastavimo, da je $\Delta = \Delta^* e^{i2\phi_k}$):

$$v_k = \sin\left(\frac{\theta_k}{2}\right) e^{i\phi_k} \quad (6)$$

$$u_k = \cos\left(\frac{\theta_k}{2}\right) \quad (7)$$

$$u_k v_k = \cos\left(\frac{\theta_k}{2}\right) \sin\left(\frac{\theta_k}{2}\right) = \frac{1}{2} \sin \theta_k \quad (8)$$

$$u_k^2 - v_k^2 = \cos^2\left(\frac{\theta_k}{2}\right) - \sin^2\left(\frac{\theta_k}{2}\right) = \cos \theta_k \quad (9)$$

skupaj dobimo (od tu naprej $\Delta := |\Delta|$):

$$\xi_k \sin(\theta_k) = \Delta \cos(\theta_k) \quad (10)$$

kar fiksira

$$\tan(\theta_k) = \frac{|\Delta_k|}{\xi_k}. \quad (11)$$

Tako iz trikotnika preberemo

$$\sin \theta_k = \frac{|\Delta_k|}{E_k} = \frac{|\Delta_k|}{\sqrt{\xi_k^2 + |\Delta_k|^2}} \quad (12)$$

$$\cos \theta_k = \frac{\xi_k}{E_k} = \frac{\xi_k}{\sqrt{\xi_k^2 + |\Delta_k|^2}} \quad (13)$$

$$(14)$$

kjer je $E_k = \pm \sqrt{\xi_k^2 + |\Delta_k|^2}$ energija kvazidelcev:

$$H = \sum_k E_k \alpha_k^\dagger \alpha_k - E_k \alpha_{-k}^\dagger \alpha_{-k} + \xi_k + \frac{|\Delta|^2}{V}. \quad (15)$$

V termičnem stanju so diagonalni delci porazdeljeni po

$$\langle \alpha_k^\dagger \alpha_k \rangle = \frac{1}{1 + \exp(E_k/T)} = \frac{1}{2} - \frac{1}{2} \tanh\left(\frac{E_k}{2T}\right). \quad (16)$$

Uporabni sta zvezi $f(-E/T) = 1 - f(E/T)$ in $1 - 2f(E/T) = \tanh(E/(2T))$. V definiciji ureditvenega parametra je

$$\langle c_k c_{-k} \rangle = \left\langle \left(u_k \alpha_k - v_k \alpha_{-k}^\dagger \right) \left(u_k \alpha_{-k} + v_k \alpha_k^\dagger \right) \right\rangle = u_k v_k \left(1 - \langle \alpha_{-k}^\dagger \alpha_{-k} \rangle - \langle \alpha_k^\dagger \alpha_k \rangle \right) = \frac{1}{2} \sin \theta_k \tanh\left(\frac{\beta E_k}{2}\right) \quad (17)$$

iz česar sledi enačba za režo pri $T > 0$ (seštevamo po k okoli E_F):

$$\Delta = \frac{V}{2} \sum_k' \sin \theta_k \tanh\left(\frac{E_k}{2T}\right) = \frac{V\Delta}{2} \sum_k \frac{\tanh\left(\frac{1}{2T} \sqrt{\Delta^2 + \xi_k^2}\right)}{\sqrt{\Delta^2 + \xi_k^2}} \quad (18)$$

1.2 Minimalni model ekscitonskega izolatorja pri končni temperaturi

Pri izolatorju za minimalni model vzamemo dve orbitali (a, b) model tesne vezi v vsaki posebej in lokalno Coulombovo interakcijo na mestu j med elektronskima gostotama v posameznih pasovih

$$H = H_0 + H_V = \sum_{i,j} \left(\epsilon_a^{(0)} \delta_{ij} + t_{ij}^{(a)} \right) c_{a,i}^\dagger c_{a,j} + \sum_{i,j} \left(\epsilon_b^{(0)} \delta_{ij} + t_{ij}^{(b)} \right) c_{b,i}^\dagger c_{b,j} + V \sum_j c_{a,j}^\dagger c_{a,j} c_{b,j}^\dagger c_{b,j} \quad (19)$$

$$= \sum_k \epsilon_{a,k} c_{a,k}^\dagger c_{a,k} + \sum_k \epsilon_{b,k} c_{b,k}^\dagger c_{b,k} + V \sum_j c_{a,j}^\dagger c_{a,j} c_{b,j}^\dagger c_{b,j} \quad (20)$$

najprej diagonaliziramo vsako orbitalo posebej, kar da gole disperzije (vezavni energiji nastavimo na $\pm G/2$) (delamo v dveh dimenzijah in mrežna konstanta $a_0 = 1$):

$$\epsilon_{a(b),k} = \pm \frac{G}{2} - 2t_{a(b)} (\cos(k_x) + \cos(k_y)), \quad (21)$$

v tem primeru je najmanjša razdalja med pasovoma $G_0 = G - 4(t_a - t_b)$ oziroma $G_0 = G - 8t_a$, če je $t_b = -t_a$. Za interakcijo spet naredimo približek povprečnega polja:

$$c_{a,j}^\dagger c_{a,j} c_{b,j}^\dagger c_{b,j} \approx \langle c_{a,j}^\dagger c_{a,j} \rangle c_{b,j}^\dagger c_{b,j} + \langle c_{b,j}^\dagger c_{b,j} \rangle c_{a,j}^\dagger c_{a,j} - \langle c_{a,j}^\dagger c_{b,j} \rangle c_{b,j}^\dagger c_{a,j} - \langle c_{b,j}^\dagger c_{a,j} \rangle c_{a,j}^\dagger c_{b,j} + \text{konst.} \quad (22)$$

od tod sledi

$$H_V^{\text{MF}} = V n_b \sum_j c_{a,j}^\dagger c_{a,j} + V n_a \sum_j c_{b,j}^\dagger c_{b,j} - V \phi \sum_j c_{b,j}^\dagger c_{a,j} - V \phi^* \sum_j c_{a,j}^\dagger c_{b,j} + NV |\phi|^2 \quad (23)$$

kjer smo definirali:

$$n_a = \langle c_{a,j}^\dagger c_{a,j} \rangle = \frac{1}{N} \sum_k \langle c_{a,k}^\dagger c_{a,k} \rangle \quad (24)$$

$$n_b = \langle c_{b,j}^\dagger c_{b,j} \rangle = \frac{1}{N} \sum_k \langle c_{b,k}^\dagger c_{b,k} \rangle \quad (25)$$

$$\phi = \langle c_{a,j}^\dagger c_{b,j} \rangle = \frac{1}{N} \sum_k \langle c_{a,k}^\dagger c_{b,k} \rangle \quad (26)$$

$$\Delta = -\frac{V}{N} \sum_k \langle c_{a,k}^\dagger c_{b,k} \rangle, \quad \Delta^* = -\frac{V}{N} \sum_k \langle c_{b,k}^\dagger c_{a,k} \rangle \quad (27)$$

$$\Delta_k = \frac{1}{N} \sum_{k'} V(k - k') \langle a_{k'}^\dagger b_{k'} \rangle, \quad V(q) = \int e^{i\mathbf{q} \cdot \mathbf{r}} V(\mathbf{r}) d\mathbf{r} \quad (28)$$

Uporabimo FT še na celotnem Hamiltonjanu

$$c_{a,j}^\dagger = \frac{1}{\sqrt{N}} \sum_k e^{i\mathbf{k} \cdot \mathbf{r}_j} c_{a,k}^\dagger \quad (29)$$

$$H^{\text{MF}} = \sum_k \left([\epsilon_{a,k} + V n_b] c_{a,k}^\dagger c_{a,k} + [\epsilon_{b,k} + V n_a] c_{b,k}^\dagger c_{b,k} + \Delta^* c_{a,k}^\dagger c_{b,k} + \Delta c_{b,k}^\dagger c_{a,k} \right) + \left(\frac{|\Delta|^2}{V} N - V N n_a n_b \right) \quad (30)$$

oziroma:

$$H^{\text{MF}} = \sum_k \begin{pmatrix} c_{a,k}^\dagger & c_{b,k}^\dagger \end{pmatrix} \begin{pmatrix} \epsilon_{a,k}^H & \Delta^* \\ \Delta & \epsilon_{b,k}^H \end{pmatrix} \begin{pmatrix} c_{a,k} \\ c_{b,k} \end{pmatrix} + \text{konst.}, \quad (31)$$

kjer definiramo Hartree energiji $\epsilon_{a(b),k}^H = \epsilon_{a(b),k} + V n_{b(a)}$. Transformacija v diagonalno / Bogoljubov bazo je podobna kot pri BCS, le da ne meša kreatorjev in anihilatorjev / ohranja število delcev:

$$\begin{pmatrix} c_{a,k} \\ c_{b,k} \end{pmatrix} = \begin{pmatrix} u_k & v_k \\ -v_k & u_k \end{pmatrix} \begin{pmatrix} \alpha_k \\ \beta_k \end{pmatrix} \quad (32)$$

od tod po enakem postopku kot prej dobimo energiji kvazidelcev

$$E_{\alpha(\beta)}(k) = \frac{\epsilon_{a,k}^H + \epsilon_{b,k}^H}{2} \pm \sqrt{\left(\frac{\epsilon_{a,k}^H - \epsilon_{b,k}^H}{2} \right)^2 + |\Delta|^2} := \eta_k \pm \sqrt{\xi_k^2 + \Delta^2} \quad (33)$$

Za bookkeeping:

$$\left(\frac{\epsilon_a^H - \epsilon_b^H}{2} \right) \sin \theta_k + \Delta \cos \theta_k = 0 \implies \tan \theta_k = \frac{-2\Delta}{\epsilon_a^H - \epsilon_b^H} = \frac{-\Delta}{\xi_k} \quad (34)$$

$$u_k v_k = -\frac{\sin \theta_k}{2}. \quad (35)$$

Za samousklajeno enačbo zdaj potrebujemo

$$\langle c_{a,k}^\dagger c_{b,k} \rangle = \langle (u_k \alpha_k^\dagger + v_k \beta_k^\dagger)(-v_k \alpha_k + u_k \beta_k) \rangle = u_k v_k \left(\langle \beta_k^\dagger \beta_k \rangle - \langle \alpha_k^\dagger \alpha_k \rangle \right) = -\frac{1}{2} \sin \theta_k (f(E_\beta) - f(E_\alpha)) \quad (36)$$

od tod

$$\Delta = -\frac{V}{N} \sum_k \langle c_{a,k}^\dagger c_{b,k} \rangle = \frac{V}{N} \sum_k \frac{\Delta}{2\sqrt{\xi_k^2 + |\Delta|^2}} (f(E_\beta) - f(E_\alpha)) \quad (37)$$

Opomba, lahko definiramo tudi $n = n_a + n_b = \text{konst.} = 1$ in $n_a - n_b =: m$ od tod sledi

$$H^{\text{MF}} = \sum_k \left(\left[\epsilon_{a,k} + \frac{V}{2} n - \frac{V}{2} m \right] c_{a,k}^\dagger c_{a,k} + \left[\epsilon_{b,k} + \frac{V}{2} n - \frac{V}{2} m \right] c_{b,k}^\dagger c_{b,k} + \Delta^* c_{a,k}^\dagger c_{b,k} + \Delta c_{b,k}^\dagger c_{a,k} \right) \quad (38)$$

od koder je v 2D Hartree gap enak:

$$G^H = G - 4(t_a - t_b) - Vm. \quad (39)$$

Note: v praksi je $m < 0$ V dveh dimenzijah se Hartree pasova srečata na

$$\mu_B = \frac{V}{2} n - \frac{t_a + t_b}{t_a - t_b} \frac{G - Vm}{2}. \quad (40)$$

Zamik je enak $V/2$, ko sta osnovna pasova simetrična in je zasedenost polovična. V primeru simetričnih pasov sta disperziji kvazidelcev

$$E_{\alpha(\beta)}(k) - V/2 = \pm \sqrt{\left(\frac{\epsilon_{a,k} - \epsilon_{b,k} - Vm}{2}\right)^2 + |\Delta|^2}. \quad (41)$$

Velja še

$$n_a = \frac{1}{N} \sum_k (u_k^2 f(E_\alpha) + v_k^2 f(E_\beta)) \quad (42)$$

$$n_b = \frac{1}{N} \sum_k (v_k^2 f(E_\alpha) + u_k^2 f(E_\beta)) \quad (43)$$

$$u_k^2 = \frac{1}{2} \left(1 + \frac{\xi_k}{\sqrt{\Delta^2 + \xi_k^2}} \right) \quad (44)$$

$$v_k^2 = \frac{1}{2} \left(1 - \frac{\xi_k}{\sqrt{\Delta^2 + \xi_k^2}} \right) \quad (45)$$

1.3 Numerika/psevdokoda

1.4 Odprti sistemi in modeliranje kopeli

Born-Markov master enačbe in Redfield: ločimo kopel in sistem,

$$\rho_S(t) = \text{Tr}_B(\rho(t)) \quad (46)$$

Hamiltonjan razcepimo na $H = H_0 + H_I$, kjer je $H_0 = H_S \otimes \mathbb{I}_B + \mathbb{I}_S \otimes H_B$ razdeljen na podсистema, za interakcijski del pa privzamemo obliko $H_I = \sum_\alpha A_\alpha \otimes B_\alpha$. V interakcijski sliki je operator za evolucijo

$$U_0(t, t_0) = e^{-iH_0(t-t_0)} \quad (47)$$

$$\rho_I(t) = U_0^\dagger(t, 0) \rho(t) U_0(t, 0) = e^{iH_0 t} \rho(t) e^{-iH_0 t} \quad (48)$$

$$H_I(t) = U_0^\dagger(t, 0) H_I U_0(t, 0) = e^{iH_0 t} H_I e^{-iH_0 t} \quad (49)$$

Von Neumannova enačba je

$$\frac{d}{dt} \rho_I(t) = -i[H_I(t), \rho_I(t)] \quad (50)$$

$$\rho_I(t) = \rho_I(0) - i \int_0^t dt' [H_I(t'), \rho_I(t')] \quad (51)$$

vstavimo nazaj v komutator in se znebimo prostosnih stopenj kopeli:

$$\frac{d}{dt} \rho_S(t) = - \int_0^t ds \text{Tr}_B ([H_I(t), [H_I(t-s), \rho_S(t) \otimes \rho_B]]) \quad (52)$$

V interakcijskem členu so operatorji A_α napisani kot lastni operatorji H_S z energijami prehodov $\omega = E_b - E_a$

$$A_\alpha(\omega) = \sum_{E_b - E_a = \omega} |a\rangle \langle a| A_\alpha |b\rangle \langle b| \quad (53)$$

To zadošča relacijam $[H_S, A_\alpha(\omega)] = -\omega A_\alpha(\omega)$, $[H_S, A_\alpha^\dagger(\omega)] = \omega A_\alpha^\dagger(\omega)$ in $e^{iH_S t} A_\alpha(\omega) e^{-iH_S t} = e^{-i\omega t} A_\alpha(\omega)$, $e^{iH_S t} A_\alpha^\dagger(\omega) e^{-iH_S t} = e^{i\omega t} A_\alpha^\dagger(\omega)$. V interakcijski sliki potem velja

$$H_I(t) = \sum_{\alpha, \omega} e^{-i\omega t} A_\alpha(\omega) \otimes B_\alpha(t) = e^{iH_B t} B_\alpha e^{-iH_B t} \quad (54)$$

evolucija sledi kot

$$\dot{\rho}_S = \sum_{\omega, \omega'} \sum_{\alpha, \beta} e^{i(\omega' - \omega)t} \Gamma_{\alpha\beta}(\omega) \left(A_\alpha(\omega) \rho_S(t) A_\beta^\dagger(\omega') - A_\beta^\dagger(\omega') A_\alpha(\omega) \rho_S(t) \right) + \text{h.c.} \quad (55)$$

kjer so prehodne matrike med stanji

$$\Gamma_{\alpha\beta}(\omega) = \int_0^\infty ds e^{i\omega s} \text{Tr}_B \left(B_\alpha^\dagger(t) B_\beta(t-s) \rho_B \right) = \frac{1}{2} \gamma_{\alpha\beta}(\omega) + i S_{\alpha\beta}(\omega) \quad (56)$$

Evolucija pa ima strukturo

$$\frac{d\rho_S}{dt} = -i[H_{LS}, \rho_S(t)] + \mathcal{D}(\rho_S(t)) \quad (57)$$

z disipatorjem

$$\mathcal{D}(\rho_S) = \sum_{\omega} \sum_{\alpha, \beta} \gamma_{\alpha\beta}(\omega) \left(A_\beta(\omega) \rho_S A_\alpha^\dagger(\omega) - \frac{1}{2} \{ A_\alpha^\dagger(\omega) A_\beta(\omega), \rho_S \} \right) \quad (58)$$

oz. v splošnem

$$\mathcal{D}(\rho_S) = \sum_k \gamma_k \left(L_k \rho_S L_k^\dagger - \frac{1}{2} \{ L_k^\dagger L_k, \rho_S \} \right) \quad (59)$$

V stacionarnem stanju velja

$$\mathcal{L}\rho_{\text{ness}} = 0. \quad (60)$$

Notes za vektorizacijo in superoperatorje:

$$\mathcal{L}u_\mu^R = \lambda_\mu u_\mu^R \quad (61)$$

$$\text{vec}(ABC) = (A \otimes C^T) \text{vec}(B) \quad (62)$$

$$\text{Tr}(A^\dagger B) = \text{vec}(A)^\dagger \text{vec}(B) \quad (63)$$

$$\frac{d}{dt} |\rho\rangle = \left(-iH \otimes \mathbb{I} + i\mathbb{I} \otimes H^T + \sum_k L_k \otimes L_k^* - \frac{1}{2} \sum_k (L_k^\dagger L_k \otimes \mathbb{I} + \mathbb{I} \otimes L_k^T L_k^*) \right) |\rho\rangle \quad (64)$$

$$\langle \hat{O} \rangle(t) = \langle \mathbb{I} | \hat{O} \otimes \mathbb{I} | \rho(t) \rangle \quad (65)$$

Šibke sklopitve in perturbacijski pristopi

$$\mathcal{L} = -i[H, \cdot] + \epsilon \mathcal{D}[\cdot], \quad \rho = \rho_0 + \epsilon \rho_1 + \epsilon^2 \rho_2 + \dots \quad (66)$$

napišemo gostotno matriko sistema v lastni bazi, siagonalno, s klasičnimi populacijami

$$\rho_0 = \sum_n p_n |n\rangle \langle n|, \quad H|n\rangle = E_n |n\rangle \quad (67)$$

kjer velja v ničtem redu (by construction)

$$\mathcal{L}\rho_0 = -i[H, \rho_0] = 0 \quad (68)$$

V prvem redu so enačbe za NESS

$$-i[H, \rho_1] + \mathcal{D}(\rho_0) = 0 \quad (69)$$

kjer je $\mathcal{D}(\rho_0) = \sum_{\mu} (L_{\mu}\rho_0 L_{\mu}^{\dagger} - \frac{1}{2}\{L_{\mu}^{\dagger}L_{\mu}, \rho_0\})$ in od tod enačba za prehode med populacijami

$$\langle n | \mathcal{D}(\rho_0) | n \rangle = \sum_{\mu} \sum_m p_m |\langle n | L_{\mu} | m \rangle|^2 - \sum_{\mu} \sum_m p_n |\langle m | L_{\mu} | n \rangle|^2 \quad (70)$$

kot klasična populacijska dinamika z $R_{nm} = \sum_{\mu} |\langle n | L_{\mu} | m \rangle|^2$

$$\dot{p}_n = \epsilon \sum_m (R_{nm} p_m - R_{mn} p_n) \quad (71)$$

Izven diagonalni elementi se razvijejo hitro na $\mathcal{O}(1)$, medtem ko se populacije razvijajo počasi na GGE manifoldu z $\mathcal{O}(\epsilon^{-1})$. Za ohranitev normalizacije mora veljati $\sum_n \dot{p}_n = 0$.

Integrabilni sistemi imajo makroskopsko število ohranjenih nabojev $[H_0, C_i] = 0$, pod unitarno časovno evolucijo se razvijejo v netermično stanje

$$\rho_{GGE} = \frac{1}{Z} e^{-\sum_i \lambda_i C_i} \quad (72)$$

$$\langle \psi_0 | C_i | \psi_0 \rangle = \text{Tr}(C_i \rho_{GGE}) \quad (73)$$

fiksira λ_i . UNitarna dinamika pripcira na GGE manifold, šibka sklopitev premika po GGE manifoldu:

$$\rho_0(t) = \rho_{\lambda(t)} = \frac{1}{Z(t)} \exp \left(- \sum_i \lambda_i(t) C_i \right) \quad (74)$$

EOM za ohranjene naboje je

$$\langle \dot{C}_i \rangle_{\lambda(\infty)} = \text{Tr}[C_i \mathcal{D}\rho] \simeq \epsilon \text{Tr}[C_i \mathcal{D}\rho_0] = 0 \quad \forall \lambda_i \quad (75)$$

Definiramo projektor na GGE manifold:

$$PX = - \sum_{ij} \frac{\partial \rho}{\partial \lambda_i} (\chi^{-1})_{ij} \text{Tr}[C_j X] \quad (76)$$

ki da dinamiko vzdolž manifolda:

$$\frac{\partial \rho_{\lambda}}{\partial t} = \sum_i \frac{\partial \rho_{\lambda}}{\partial \lambda_i} \dot{\lambda}_i = \epsilon P \mathcal{D}(\rho_{\lambda}) = -\epsilon \sum_{ij} \frac{\partial \rho_{\lambda}}{\partial \lambda_i} (\chi^{-1})_{ij} \text{Tr}[C_j \mathcal{D}(\rho_{\lambda})] \quad (77)$$

od koder sledi evolucija za koeficiente/potenciale λ_i :

$$\dot{\lambda}_i = -\epsilon \sum_j (\chi^{-1})_{ij} \text{Tr}[C_j \mathcal{D}\rho_{\lambda(t)}] \quad (78)$$

s susceptibilnostmi $\chi_{ij} = \langle C_i C_j \rangle - \langle C_i \rangle \langle C_j \rangle = -\text{Tr} \left[C_i \frac{\partial \rho}{\partial \lambda_j} \right]$.

2 Ekscitonski izolatorji in sklopitev s kopelmi

$$\text{Tr} \left[f \frac{1}{2} \{\rho, A\} \right] = \text{Tr} \left[\frac{1}{2} f \rho A + \frac{1}{2} f A \rho \right] = \frac{1}{2} \text{Tr} [A f \rho + f A \rho] \quad (79)$$

$$= \frac{1}{2} \text{Tr} [(A f + f A) \rho] = \frac{1}{2} \langle A f + f A \rangle \quad (80)$$

2.1 Osnovni modeli kopeli

2.2 Limita neskončne temperature

2.3 Fononske sklopitve

Začnemo z modelom, ki ga modificiram od prej za interband sklopitve:

$$H(\{\mathbf{u}\}) = \underbrace{\sum_{i,\alpha,\beta} \epsilon_{\alpha\beta}(\mathbf{u}_i) c_{\alpha,i}^\dagger c_{\beta,i}}_{\text{on-site in CEF}} + \underbrace{\sum_{\langle i,j \rangle, \alpha, \beta} t_{ij}^{(\alpha\beta)}(\mathbf{r}_{ij}) c_{\alpha,i}^\dagger c_{\beta,j}}_{\text{hopping, med orbitalami}} + \underbrace{V \sum_j n_{a,j} n_{b,j}}_{\text{Coulomb}} \quad (81)$$

kjer privzamemo:

$$t_{ij}^{(aa)}(\mathbf{r}_{ij}) \approx t_{ij}^{(a)0} + \alpha_t^{(a)} \hat{\mathbf{R}}_{ij}^0 \cdot (\mathbf{u}_j - \mathbf{u}_i) \quad (82)$$

$$t_{ij}^{(bb)}(\mathbf{r}_{ij}) \approx t_{ij}^{(b)0} + \alpha_t^{(b)} \hat{\mathbf{R}}_{ij}^0 \cdot (\mathbf{u}_j - \mathbf{u}_i) \quad (83)$$

$$t_{ij}^{(ab)}(\mathbf{r}_{ij}) \approx \underbrace{0}_{\text{ni hoppinga v nedeform. mreži}} + \alpha_t^{(ab)} \cdot (\mathbf{u}_j - \mathbf{u}_i) \quad (84)$$

$$\alpha_t^{(ab)} = \nabla_{\mathbf{r}} t_{ij}^{(ab)} \Big|_{\mathbf{R}_{ij}^0} \neq 0 \quad (85)$$

2.4 Fotonske sklopitve v resonatorju

Literatura

- [1] Monique Combescot in Shiue-Yuan Shiao. *Excitons and cooper pairs: two composite bosons in many-body physics*. Oxford graduate texts. Oxford: Oxford university press, 2016. ISBN: 978-0-19-875373-5.
- [2] D. Jérôme, T. M. Rice in W. Kohn. “Excitonic Insulator”. V: *Physical Review* 158.2 (10. jun. 1967), str. 462–475. ISSN: 0031-899X. DOI: [10.1103/PhysRev.158.462](https://doi.org/10.1103/PhysRev.158.462). URL: <https://link.aps.org/doi/10.1103/PhysRev.158.462> (pridobljeno 27. 6. 2026).
- [3] A. Asenjo-Garcia in sod. “Exponential Improvement in Photon Storage Fidelities Using Subradiance and “Selective Radiance” in Atomic Arrays”. V: *Physical Review X* 7.3 (3. avg. 2017), str. 031024. ISSN: 2160-3308. DOI: [10.1103/PhysRevX.7.031024](https://doi.org/10.1103/PhysRevX.7.031024). URL: <http://link.aps.org/doi/10.1103/PhysRevX.7.031024> (pridobljeno 27. 6. 2026).
- [4] Gianluca Lagnese in sod. *Neural network modeling of many-body super- and sub-radiant dynamics*. 6. maj 2026. DOI: [10.48550/arXiv.2605.04640](https://doi.org/10.48550/arXiv.2605.04640). arXiv: [2605.04640](https://arxiv.org/abs/2605.04640)[quant-ph]. URL: <http://arxiv.org/abs/2605.04640> (pridobljeno 27. 6. 2026).
- [5] Kwangrae Kim in sod. “Direct observation of excitonic instability in Ta2NiSe5”. V: *Nature Communications* 12.1 (30. mar. 2021), str. 1969. ISSN: 2041-1723. DOI: [10.1038/s41467-021-22133-z](https://doi.org/10.1038/s41467-021-22133-z). arXiv: [2007.08212](https://arxiv.org/abs/2007.08212)[cond-mat.str-el]. URL: <http://arxiv.org/abs/2007.08212> (pridobljeno 27. 6. 2026).
- [6] Yuta Murakami in sod. “Collective Modes in Excitonic Insulators: Effects of Electron-Phonon Coupling and Signatures in Optical Response”. V: *Physical Review B* 101.19 (11. maj 2020), str. 195118. ISSN: 2469-9950, 2469-9969. DOI: [10.1103/PhysRevB.101.195118](https://doi.org/10.1103/PhysRevB.101.195118). arXiv: [2003.10789](https://arxiv.org/abs/2003.10789)[cond-mat.str-el]. URL: <http://arxiv.org/abs/2003.10789> (pridobljeno 28. 6. 2026).
- [7] Tatsuya Kaneko in Yukinori Ohta. “A New Era of Excitonic Insulators”. V: *Journal of the Physical Society of Japan* 94.1 (15. jan. 2025), str. 012001. ISSN: 0031-9015, 1347-4073. DOI: [10.7566/JPSJ.94.012001](https://doi.org/10.7566/JPSJ.94.012001). arXiv: [2411.10985](https://arxiv.org/abs/2411.10985)[cond-mat.str-el]. URL: <http://arxiv.org/abs/2411.10985> (pridobljeno 28. 6. 2026).
- [8] Daniel Arovas. “Lecture Notes on Condensed Matter Physics (A Work in Progress)”. V: ().
- [9] B. Hülse in sod. “Phase diagram of the excitonic insulator”. V: *Physica B: Condensed Matter* 378-380 (maj 2006), str. 267–268. ISSN: 09214526. DOI: [10.1016/j.physb.2006.01.099](https://doi.org/10.1016/j.physb.2006.01.099). URL: <https://linkinghub.elsevier.com/retrieve/pii/S092145260600130X> (pridobljeno 9. 8. 2026).
- [10] V. Apinyan in T. K. Kopeč. “Phase coherence and spectral functions in the two-dimensional excitonic systems”. V: *Physica B: Condensed Matter* 473 (sep. 2015), str. 75–92. ISSN: 09214526. DOI: [10.1016/j.physb.2015.05.027](https://doi.org/10.1016/j.physb.2015.05.027). arXiv: [1311.1645](https://arxiv.org/abs/1311.1645)[cond-mat.str-el]. URL: <http://arxiv.org/abs/1311.1645> (pridobljeno 9. 8. 2026).

- [11] D. I. Golosov. “Excitonic Insulator and the Extended Falicov–Kimball Model Away from Half-Filling”. V: *Physical Review B* 112.24 (26. dec. 2025), str. 245161. ISSN: 2469-9950, 2469-9969. DOI: [10.1103/19zh-ldpv](https://doi.org/10.1103/19zh-ldpv). arXiv: [2507.13967](https://arxiv.org/abs/2507.13967) [cond-mat.str-el]. URL: <http://arxiv.org/abs/2507.13967> (pridobljeno 9. 8. 2026).
- [12] Farkašovský. “DMRG study of exciton condensation in the extended Falicov–Kimball model”. V: *Condensed Matter Physics* 23.4 (dec. 2020), str. 43709. ISSN: 1607324X, 22249079. DOI: [10.5488/CMP.23.43709](https://doi.org/10.5488/CMP.23.43709). URL: <http://www.icmp.lviv.ua/journal/zbirnyk.104/43709/abstract.html> (pridobljeno 9. 8. 2026).
- [13] Francesco Campaioli, Jared H. Cole in Harini Hapuarachchi. “Quantum Master Equations: Tips and Tricks for Quantum Optics, Quantum Computing, and Beyond”. V: *PRX Quantum* 5.2 (10. jun. 2024), str. 020202. ISSN: 2691-3399. DOI: [10.1103/PRXQuantum.5.020202](https://doi.org/10.1103/PRXQuantum.5.020202). URL: <https://link.aps.org/doi/10.1103/PRXQuantum.5.020202> (pridobljeno 12. 8. 2026).
- [14] Iris Ulčakar in Zala Lenarčič. “Generalized Gibbs ensembles in weakly interacting dissipative systems and digital quantum computers”. V: *SciPost Physics* 19.3 (4. sep. 2025), str. 068. ISSN: 2542-4653. DOI: [10.21468/SciPostPhys.19.3.068](https://doi.org/10.21468/SciPostPhys.19.3.068). URL: <https://scipost.org/10.21468/SciPostPhys.19.3.068> (pridobljeno 13. 8. 2026).
- [15] E. Ilievski in sod. “Complete Generalized Gibbs Ensembles in an Interacting Theory”. V: *Physical Review Letters* (2015). DOI: [10.1103/PhysRevLett.115.157201](https://doi.org/10.1103/PhysRevLett.115.157201). URL: <https://sci-hub.box/10.1103/PhysRevLett.115.157201> (pridobljeno 13. 8. 2026).
- [16] Marcos Rigol in sod. “Relaxation in a Completely Integrable Many-Body Quantum System: An Ab Initio Study of the Dynamics of the Highly Excited States of Lattice Hard-Core Bosons”. V: *Physical Review Letters* 98.5 (1. feb. 2007), str. 050405. ISSN: 0031-9007, 1079-7114. DOI: [10.1103/PhysRevLett.98.050405](https://doi.org/10.1103/PhysRevLett.98.050405). arXiv: [cond-mat/0604476](https://arxiv.org/abs/cond-mat/0604476). URL: <http://arxiv.org/abs/cond-mat/0604476> (pridobljeno 13. 8. 2026).
- [17] Scott Davidson in sod. “The dark side of energy transport along excitonic wires: On-site energy barriers facilitate efficient, vibrationally mediated transport through optically dark subspaces”. V: *The Journal of Chemical Physics* (2020). DOI: [10.1063/5.0023702](https://doi.org/10.1063/5.0023702). URL: <https://sci-hub.box/10.1063/5.0023702> (pridobljeno 16. 8. 2026).
- [18] Florian Lange, Zala Lenarčič in Achim Rosch. “Pumping approximately integrable systems”. V: *Nature Communications* 8.1 (9. jun. 2017), str. 15767. ISSN: 2041-1723. DOI: [10.1038/ncomms15767](https://doi.org/10.1038/ncomms15767). arXiv: [1608.03563](https://arxiv.org/abs/1608.03563) [cond-mat.str-el]. URL: <http://arxiv.org/abs/1608.03563> (pridobljeno 16. 8. 2026).
- [19] Cristian Zanoci, Yongchan Yoo in Brian Swingle. “Thermalization at Low Temperatures via Weakly-Damped Multi-Site Baths”. V: *Physical Review B* 108.3 (31. jul. 2023), str. 035156. ISSN: 2469-9950, 2469-9969. DOI: [10.1103/PhysRevB.108.035156](https://doi.org/10.1103/PhysRevB.108.035156). arXiv: [2304.08525](https://arxiv.org/abs/2304.08525) [cond-mat.str-el]. URL: <http://arxiv.org/abs/2304.08525> (pridobljeno 17. 8. 2026).
- [20] Ana Štuhec. “Termoelektrični pojav v ekscitonskih izolatorjih”. V: ().
- [21] Andrew J. Daley. “Quantum trajectories and open many-body quantum systems”. V: *Advances in Physics* 63.2 (4. mar. 2014), str. 77–149. ISSN: 0001-8732, 1460-6976. DOI: [10.1080/00018732.2014.933502](https://doi.org/10.1080/00018732.2014.933502). URL: <http://www.tandfonline.com/doi/abs/10.1080/00018732.2014.933502> (pridobljeno 19. 8. 2026).
- [22] Elahe Davari in Mehdi Kargarian. “Probing the collective excitations of excitonic insulators in an optical cavity”. V: *Physical Review B* 112.16 (17. okt. 2025), str. 165124. ISSN: 2469-9950, 2469-9969. DOI: [10.1103/zqhl-kncx](https://doi.org/10.1103/zqhl-kncx). arXiv: [2501.04736](https://arxiv.org/abs/2501.04736) [cond-mat.mes-hall]. URL: <http://arxiv.org/abs/2501.04736> (pridobljeno 25. 8. 2026).
- [23] E. T. Jaynes in F. W. Cummings. “Comparison of quantum and semiclassical radiation theories with application to the beam maser”. V: *Proceedings of the IEEE* (1963). DOI: [10.1109/PROC.1963.1664](https://doi.org/10.1109/PROC.1963.1664). URL: <https://sci-hub.box/10.1109/PROC.1963.1664> (pridobljeno 25. 8. 2026).
- [24] Anneswa Paul in Upendra Harbola. *Light-matter interaction inside an optical cavity: A perspective*. 27. apr. 2025. DOI: [10.48550/arXiv.2504.14266](https://doi.org/10.48550/arXiv.2504.14266). arXiv: [2504.14266](https://arxiv.org/abs/2504.14266) [quant-ph]. URL: <http://arxiv.org/abs/2504.14266> (pridobljeno 25. 8. 2026).
- [25] Arkajit Mandal in sod. *Theoretical Advances in Polariton Chemistry and Molecular Cavity Quantum Electrodynamics*. 7. dec. 2022. DOI: [10.26434/chemrxiv-2022-g9lr7](https://doi.org/10.26434/chemrxiv-2022-g9lr7). URL: <https://chemrxiv.org/doi/full/10.26434/chemrxiv-2022-g9lr7> (pridobljeno 25. 8. 2026).
- [26] Jiajun Li in sod. “Electromagnetic coupling in tight-binding models for strongly correlated light and matter”. V: *Physical Review B* 101.20 (21. maj 2020), str. 205140. ISSN: 2469-9950, 2469-9969. DOI: [10.1103/PhysRevB.101.205140](https://doi.org/10.1103/PhysRevB.101.205140). arXiv: [2001.09726](https://arxiv.org/abs/2001.09726) [cond-mat.str-el]. URL: <http://arxiv.org/abs/2001.09726> (pridobljeno 25. 8. 2026).

- [27] Simon B. Jäger in sod. “Lindblad master equations for quantum systems coupled to dissipative bosonic modes”. V: *Physical Review Letters* 129.6 (2. avg. 2022), str. 063601. ISSN: 0031-9007, 1079-7114. DOI: [10.1103/PhysRevLett.129.063601](https://doi.org/10.1103/PhysRevLett.129.063601). arXiv: [2203.03302\[quant-ph\]](https://arxiv.org/abs/2203.03302). URL: <http://arxiv.org/abs/2203.03302> (pridobljeno 25. 8. 2026).
- [28] Gerald Dunne in Roman Jackiw. “Peireles Substitution” and Chern-Simons Quantum Mechanics”. V: *Nuclear Physics B - Proceedings Supplements* 33.3 (nov. 1993), str. 114–118. ISSN: 09205632. DOI: [10.1016/0920-5632\(93\)90376-H](https://doi.org/10.1016/0920-5632(93)90376-H). arXiv: [hep-th/9204057](https://arxiv.org/abs/hep-th/9204057). URL: <http://arxiv.org/abs/hep-th/9204057> (pridobljeno 26. 8. 2026).
- [29] Florian Lange, Zala Lenarčič in Achim Rosch. “Time-dependent generalized Gibbs ensembles in open quantum systems”. V: *Physical Review B* 97.16 (26. apr. 2018), str. 165138. ISSN: 2469-9950, 2469-9969. DOI: [10.1103/PhysRevB.97.165138](https://doi.org/10.1103/PhysRevB.97.165138). arXiv: [1801.07646\[cond-mat.str-el\]](https://arxiv.org/abs/1801.07646). URL: <http://arxiv.org/abs/1801.07646> (pridobljeno 2. 9. 2026).
- [30] Jonathan B. Curtis in sod. “Cavity Quantum Eliashberg Enhancement of Superconductivity”. V: *Physical Review Letters* 122.16 (26. apr. 2019), str. 167002. ISSN: 0031-9007, 1079-7114. DOI: [10.1103/PhysRevLett.122.167002](https://doi.org/10.1103/PhysRevLett.122.167002). URL: <https://link.aps.org/doi/10.1103/PhysRevLett.122.167002> (pridobljeno 10. 9. 2026).
- [31] Mona H. Kalthoff in sod. “Nonequilibrium phase transition in a driven-dissipative quantum antiferromagnet”. V: *Physical Review Research* 4.2 (12. maj 2022), str. 023115. ISSN: 2643-1564. DOI: [10.1103/PhysRevResearch.4.023115](https://doi.org/10.1103/PhysRevResearch.4.023115). URL: <https://link.aps.org/doi/10.1103/PhysRevResearch.4.023115> (pridobljeno 16. 9. 2026).